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Average Cayley genus for Cayley maps with dihedral groups

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**AVERAGE CAYLEY GENUS FOR CAYLEY
MAPS WITH DIHEDRAL GROUPS**

by

Jamie Keith Langille

**Bachelor of Science
University of Nevada, Las Vegas
1998**

**A thesis submitted in partial fulfillment
of the requirements for the**

**Master of Science Degree
Department of Mathematical Sciences
College of Sciences**

**Graduate College
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Master of Science

Examination Committee Chair

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ABSTRACT

Average Cayley Genus for Cayley Maps with Dihedral Groups

by

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Let Γ be a finite group and let Δ be a generating set for Γ . A Cayley map associated with Γ and Δ is an oriented 2-cell embedding of the Cayley graph $G_\Delta(\Gamma)$ such that the rotation of arcs emanating from each vertex is determined by a unique cyclic permutation of generators and their inverses. A formula for the average Cayley genus is known for the dihedral group with generating set consisting of all the reflections. However, the known formula involves sums of certain coefficients of a generating function and its format does not specifically indicate the Cayley genus distribution. We determine a simplified formula for this average Cayley genus as well as provide improved understanding of the Cayley genus distribution.

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CHAPTER 1

INTRODUCTION AND PRELIMINARIES

A graph is a finite nonempty set of objects called vertices together with a set of unordered pairs of distinct vertices called edges. The number of vertices is the order of the graph and the number of edges is the size of the graph. A question that arises is, is it possible to draw (or embed) a graph in the plane so that none of its edges cross? These types of graphs are called planar graphs. Through stereographic projection, embedding a graph in the plane is equivalent to embedding that same graph on the sphere, which we denote by S_0 . However, there are many graphs that cannot be embedded on the sphere so that none of its edges cross. We define a surface to be a closed orientable 2-manifold, which can be thought of as a sphere with handles (or equivalently, holes). The number of handles (or holes) is the genus of the surface. For an integer $k \geq 0$, let S_k denote the surface of genus k .

We now formalize the concept of embedding. A graph G embeds on a surface S_k if there exists a homeomorphism of G as a finite 1-complex in $\mathbb{R}^3$ with a subspace of S_k . If every region of an embedding of G on S_k is homeomorphic to an open disk, then G is 2-cell embedded on S_k and we write $G \triangleleft S_k$. In this work, we require all embeddings to be 2-cell. Figure 1.A shows an example of a graph that is drawn on the torus. The left figure is not 2-cell embedded on the torus since a closed curve can be drawn in the outer

region as shown that cannot be contracted to a point, whereas the right figure is the same graph redrawn so that it is now 2-cell embedded on the torus.

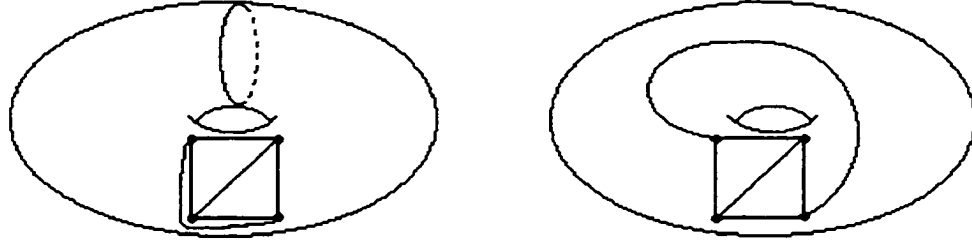


Figure 1.A Example of Non 2-Cell Embedding and 2-Cell Embedding.

What we would like to do is to be able to tell which surfaces we are able to 2-cell embed a graph on. To find which surface a graph is embedded on, we often make use of Euler's identity.

Theorem 1.1 If G is a connected graph with order p and size q that is 2-cell embedded on S_k with r regions, then

$$p - q + r = 2 - 2k,$$

where the value $2 - 2k$ is called the Euler characteristic of S_k .

Let G be a connected graph. Then the genus $\gamma(G)$ of G is the minimum non-negative integer k such that G embeds on S_k . A rotation embedding scheme $\wp$ is a collection of cyclic permutations $\rho_v : N(v) \rightarrow N(v)$, one for each $v \in V(G)$, where $V(G)$ is the set of vertices of G and $N(v)$ denotes the neighborhood of v . It is well

known (see Edmonds [4]) that the 2-cell embeddings of a connected graph G are in one-to-one correspondence with the rotation schemes of G .

Theorem 1.2 Let G be a connected graph with $V(G) = \{1, 2, \dots, p\}$. If $G \triangleleft S_k$, then this 2-cell embedding uniquely determines a rotation scheme

$$\wp = \{\rho_i : N(i) \rightarrow N(i) \mid 1 \leq i \leq p\}.$$

Conversely, such a rotation scheme uniquely determines a 2-cell embedding of G on some surface.

It is not difficult to count the number of distinct rotation schemes for a given labeled connected graph G . Indeed for a connected graph G with $V(G) = \{1, 2, \dots, p\}$, there

are $\prod_{i=1}^p (\deg(i) - 1)!$ such schemes and due to Theorem 1.2, there are then

$\prod_{i=1}^p (\deg(i) - 1)!$ many labeled 2-cell embeddings of G . Since there are a finite number of 2-cell embeddings of a given graph, there is a surface of maximum genus on which the graph can be 2-cell embedded. We now define the maximum genus $\gamma_M(G)$ of a graph G to be the maximum integer k such that $G \triangleleft S_k$. So now we have a minimum value and a maximum value, but what of the intermediate values? Duke [3] established the intermediate value theorem for genus distribution, typically referred to as Duke's Theorem.

Theorem 1.3 There is a 2-cell embedding of a connected graph G on S_k if and only if

$$\gamma(G) \leq k \leq \gamma_M(G).$$

One of the major areas of research has been the study of all 2-cell embeddings of a labeled connected graph G and, in particular, the enumeration of the embeddings of G on a given surface and the determination of the average genus of G .

For example, we shall look at the three non-labeled 2-cell embeddings of the complete graph K_4 (See Figure 1.B). So what we would like to do now is to enumerate the labeled embeddings. Figure 1.B.a is K_4 drawn on the plane and it has two labeled embeddings. Figure 1.B.b and 1.B.c are K_4 drawn on the torus (using the planar representation of a torus). Labeling these embeddings, figure 1.B.b has eight labeled embeddings and figure 1.B.c has six labeled embeddings. So there is a total of sixteen labeled 2-cell embeddings with two on S_0 and fourteen on S_1 . Hence, the average genus for K_4 is $\frac{2(0)+14(1)}{16} = \frac{14}{16}$. In general, it is a difficult problem to determine the average genus for classes of graphs.

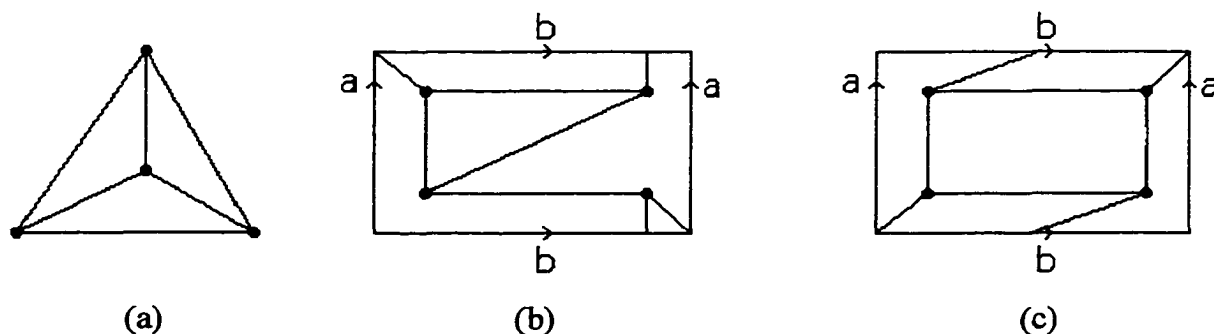


Figure 1.B The Three Embeddings of K_4 , (a) on the Plane and (b) and (c) on the Torus.

If $\wp$ is a rotation scheme for G , then the ordered pair $(G, \wp)$ is called a map and we say that the genus $g(G, \wp)$ of the map $(G, \wp)$ is k if $\wp$ determines a 2-cell embedding of G on S_k . Thus, $\gamma(G) = \min_{\wp} g(G, \wp)$ and $\gamma_M(G) = \max_{\wp} g(G, \wp)$.

The embeddings having non-trivial symmetries are often represented by smaller graph embeddings, called voltage graph embeddings. For our purposes, we will concentrate on the index one voltage graphs (those with exactly one vertex). The reason to focus on these particular voltage graphs is that our primary interest is in studying certain embeddings of Cayley graphs and these are precisely the coverings of index one voltage graphs.

Let Γ be a finite group and Δ be a generating set for Γ such that the identity $e \notin \Delta$. Also let $\Delta^{-1} = \{\delta^{-1} \mid \delta \in \Delta\}$ and $\Delta^* = \Delta \cup \Delta^{-1}$. Furthermore, let Δ be chosen so that if $\delta \in \Delta \cap \Delta^{-1}$, then $\delta^2 = e$. That is, if δ is chosen as a generator, then δ^{-1} is not chosen, unless $\delta^2 = e$ (δ is its own inverse). The Cayley graph $G_{\Delta}(\Gamma)$ is that graph whose vertex set is Γ and edge set is $\{\{x, x\delta\} \mid x \in \Gamma, \delta \in \Delta^*\}$. After defining a Cayley graph, we can now define a Cayley map. Let Γ be a group and Δ be a generating set for Γ with the usual properties, i.e. $e \notin \Delta$ and if $\delta \in \Delta \cap \Delta^{-1}$, then $\delta^2 = e$ (whenever we consider a generating set, we will always require that it satisfy these conditions). For a cyclic permutation $\rho: \Delta^* \rightarrow \Delta^*$, the Cayley map (Γ, Δ, ρ) is the map $(G_{\Delta}(\Gamma), \wp)$ where $\wp = \{\rho_x \mid x \in \Gamma\}$ is the rotation scheme for $G_{\Delta}(\Gamma)$ such that $\rho_x(y) = x\rho(x^{-1}y)$ for each $x \in \Gamma$ and each $y \in N(x)$ (see figure 1.C). In other words, a Cayley map is a 2-cell

embedding of a Cayley graph in which each vertex rotation ρ_x is determined by the same cyclic ordering of the elements of Δ^* .

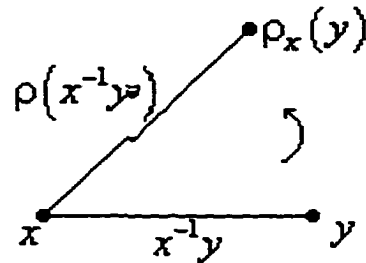


Figure 1.C Vertex Rotation in a Cayley Map.

As said earlier, Cayley maps are coverings of index one voltage graphs. Let K be a pseudograph of order 1. So K consists of one vertex and a finite number of loops. If e is a directed loop of K then we denote the reverse direction of e by e^{-1} . Let K^* be the set of all the directed loops of K and their inverses, that is, $K^* = \{e, e^{-1} \mid e \in E(K)\}$. Now let Γ be a group and let $\phi: K^* \rightarrow \Gamma$ be a one-to-one function satisfying $\phi(e^{-1}) = (\phi(e))^{-1}$ for every $e \in K^*$. Then the ordered triple (K, Γ, ϕ) is an example of a voltage graph. More specifically, (K, Γ, ϕ) is an index one voltage graph since it has order 1. The index of a voltage graph is simply the order of the pseudograph used. When the group Γ and function ϕ are unambiguous, we say simply that K is the voltage graph, where now the group and function are understood.

We now describe the covering of a voltage graph. Let Γ be a group with generating set Δ and let $\Delta^* = \Delta \cup \Delta^{-1}$. Let (K, Γ, ϕ) be an index one voltage graph such that

$\phi(E(K)) = \Delta$. That is, ϕ assigns a generator to each directed loop of K . Now, let (K, Γ, ϕ) be 2-cell embedded on a surface. The cyclic ordering of the labeled arcs emanating from the one vertex gives a cyclic permutation $\rho: \Delta^* \rightarrow \Delta^*$. This voltage graph embedding determines a 2-cell embedding of the Cayley graph $G_\Delta(\Gamma)$ in some surface and this embedding of $G_\Delta(\Gamma)$ is the Cayley map (Γ, Δ, ρ) . The Cayley graph $G_\Delta(\Gamma)$ is called the covering graph of the voltage graph (K, Γ, ϕ) and the embedding of (K, Γ, ϕ) is said to lift to the embedding of $G_\Delta(\Gamma)$. We say the embedding of $G_\Delta(\Gamma)$ is the covering embedding of the voltage graph embedding. However, in voltage graphs, a problem arises when generators of order 2 are used. A directed loop of order two produces a pair of multiple edges in the covering. To avoid this, in voltage graphs, a generator of order 2 is represented by a half-edge.

A voltage graph can be used to find the number of regions and the structure of the regions in the covering embedding. A walk w in an index one voltage graph (K, Γ, ϕ) is a sequence $f_1, f_2, \dots, f_m$ of elements of K^* . Define $\phi(w) = \prod_{i=1}^m \phi(f_i)$. Thus, $\phi(w)$ is an element of Γ . Now let R be a region of a 2-cell embedding of (K, Γ, ϕ) , and let $w: f_1, f_2, \dots, f_m$ denote the walk in K that consists of the oriented boundary of R . Then $\phi(w)$ is called the boundary element of R and $\text{ord}(\phi(w))$ is the order of the boundary element of R . This leads to a result by Gross and Alpert [5].

Theorem 1.4 If R is a k -gon in a 2-cell embedding of an index one voltage graph (K, Γ, ϕ) and s is the order of the boundary element of R , then R lifts to $\frac{|\Gamma|}{s}$ ks -gons in the covering of (K, Γ, ϕ) .

This result holds for general voltage graphs, but again, we are specializing to the case of the index one voltage graphs.

If a given Cayley map (Γ, Δ, ρ) determines a 2-cell embedding of $G_\Delta(\Gamma)$ in S_k , then k is called the genus $g(\Gamma, \Delta, \rho)$ of the Cayley map (Γ, Δ, ρ) . The Cayley genus $\gamma(\Gamma, \Delta)$ is defined as $\gamma(\Gamma, \Delta) = \min_{\rho} g(\Gamma, \Delta, \rho)$ and the maximum Cayley genus $\gamma_M(\Gamma, \Delta)$ is defined by $\gamma_M(\Gamma, \Delta) = \max_{\rho} g(\Gamma, \Delta, \rho)$. Since each Cayley map (Γ, Δ, ρ) is a 2-cell embedding of the Cayley graph $G_\Delta(\Gamma)$, we obtain the inequality $\gamma(G_\Delta(\Gamma)) \leq \gamma(\Gamma, \Delta) \leq \gamma_M(\Gamma, \Delta) \leq \gamma_M(G_\Delta(\Gamma))$. The value that we concentrate on is the average Cayley genus. The average Cayley genus $\bar{\gamma}(\Gamma, \Delta)$ is the average of the genera of all Cayley maps for some group Γ with fixed generating set Δ .

So far, we have been using Γ as an arbitrary group. What we shall do now is specialize to a particular group. We focus our attention on the dihedral group D_m of order $2m$, where $m \geq 3$. The presentation we use is $D_m = \langle x, y \mid x^m = y^2 = (xy)^2 = e \rangle$. Thus, $D_m = \{e, x, x^2, \dots, x^{m-1}, y, xy, x^2y, \dots, x^{m-1}y\}$ where $e, x, x^2, \dots, x^{m-1}$ represent the rotations and $y, xy, x^2y, \dots, x^{m-1}y$ are the reflections. Let

$\Delta = \{y, xy, x^2y, \dots, x^{m-1}y\}$. The Cayley graph $G_\Delta(D_m)$ is $K_{m,m}$, the complete bipartite graph. Since $(x^k y)^2 = e$ for all $k \in \mathbb{Z}$, we get that $yx^k = x^{m-k}y = x^{-k}y$. When m is odd, the genus for the Cayley maps (D_m, Δ, ρ) is constant, as was shown in [9].

Theorem 1.5 Let $m = 2n+1 \geq 3$. If Δ is the generating set consisting of the reflections of the dihedral group D_m , then $g(D_m, \Delta, \rho) = n(2n-1)$ for every cyclic permutation $\rho: \Delta \rightarrow \Delta$. Therefore the average Cayley genus $\bar{\gamma}(D_m, \Delta, \rho) = n(2n-1)$.

So each Cayley map (D_{2n+1}, Δ, ρ) embeds $K_{2n+1, 2n+1}$ on a surface of genus $n(2n-1)$. So the Cayley genus and maximum Cayley genus are equal for this group and generating set, meaning $\gamma(D_{2n+1}, \Delta) = \gamma_M(D_{2n+1}, \Delta) = n(2n-1)$. As a result of the genus being constant when m is an odd integer, we will concentrate on m being an even integer. So $D_m = D_{2n}$ for $n \geq 2$. Again, we take Δ as the generating set consisting of all the reflections, that is, $\Delta = \{y, xy, x^2y, \dots, x^{2n-1}y\}$ and so the Cayley graph is $G_\Delta(D_{2n}) = K_{2n, 2n}$.

Next, we need to introduce generating functions. A generating function, as mentioned in Brualdi [2], can be thought of as an object, which through manipulation, allows us to count a number of possibilities for a problem. Also, Brualdi indicates that generating functions can be thought of as Taylor series (power series expansions) of infinitely differentiable functions. By finding the function and the Taylor series, the coefficients give the solution. If we let $h_0, h_1, h_2, \dots, h_n, \dots$ be an infinite sequence of

numbers, then its generating function is the infinite series

$$g(x) = h_0 + h_1x + h_2x^2 + \dots + h_nx^n + \dots$$

So, for an example of a generating function, the binomial coefficients in the expansion of $(a+b)^m$ are $\binom{m}{0}, \binom{m}{1}, \binom{m}{2}, \dots, \binom{m}{m}$. The generating function for these coefficients is $g_m(x) = \binom{m}{0} + \binom{m}{1}x + \binom{m}{2}x^2 + \dots + \binom{m}{m}x^m$, or $g_m(x) = (1+x)^m$ by the binomial theorem.

Finally, we need to introduce the idea of partitions of integers (see Andrews [1], for example). A partition of a positive integer n is a finite non-increasing sequence of positive integers $\lambda_1, \lambda_2, \dots, \lambda_r$ such that $\sum_{i=1}^r \lambda_i = n$. The numbers λ_i , for $1 \leq i \leq r$ are called the parts of the partition. So as an example, let's consider the partitions of 6. The only partition of 6 with six parts is $1+1+1+1+1+1$. The only partition of 6 with five parts is given by $2+1+1+1+1$. There are two partitions of 6 with four parts, $3+1+1+1$ and $2+2+1+1$. The three partitions of 6 into three parts are $4+1+1$, $3+2+1$ and $2+2+2$. We can partition 6 into two parts three ways, namely $5+1$, $4+2$ and $3+3$. Finally, 6 is the only way to partition 6 into one part. Partitions can also be restricted. For example, the number of parts can be restricted as well as the size of each part. For our work with Cayley maps for dihedral groups, it will be useful to know the number of partitions of each of the integers $\binom{n}{2}, \binom{n}{2}+1, \binom{n}{2}+2, \dots, 3\binom{n}{2}$ into exactly $n-1$ unequal parts with no part greater than $2n-1$.

Let k and j be positive integers with $k \leq j$. Then from Riordan [8], we have a generating function $u_j(t, k)$ for the number of partitions having k unequal parts with no part greater than j . That generating function is as follows:

$$u_j(t, k) = \begin{cases} t^{\binom{k+1}{2}} \frac{(1-t^j)(1-t^{j-1})\dots(1-t^{j-k+1})}{(1-t)(1-t^2)\dots(1-t^k)} & \text{if } k < j \\ t^{\binom{j+1}{2}} & \text{if } k = j \end{cases}.$$

In Riordan, this generating function appears as an exercise. To see where it comes from, recall what we are trying to do. We wish to enumerate the number of partitions with k unequal parts, no part greater than j . The function for this restricted partition is given by $G_j(t, a) = (1+at)(1+at^2)\dots(1+at^j)$. Since we want unequal parts, the general factor is $1+at^i$. The term at^i in this general factor is the indicator for one occurrence of i . Since we want no part greater than j , our last factor is $1+at^j$. So the exponent of a keeps track of how many parts are being used. Thus, $G_j(t, a) = (1+at)(1+at^2)\dots(1+at^j) = \sum_{k=0}^j u_j(t, k) a^k$, where $u_j(t, k)$ is the generating function we are looking for. We define $u_j(t, 0) = 1$, the empty partition. Start by considering $G_j(t, ta) = (1+at^2)(1+at^3)\dots(1+at^{j+1})$ so that

$$(1+at)G_j(t, ta) = G_j(t, a)(1+at^{j+1}),$$

or,

$$(1+at) \sum_{k=0}^j u_j(t, k) t^k a^k = (1+at^{j+1}) \sum_{k=0}^j u_j(t, k) a^k,$$

or by distributing,

$$\sum_{k=0}^j u_j(t, k) t^k a^k + \sum_{k=0}^j u_j(t, k) t^{k+1} a^{k+1} = \sum_{k=0}^j u_j(t, k) a^k + \sum_{k=0}^j u_j(t, k) t^{j+1} a^{k+1}$$

so that

$$\sum_{k=0}^j (t^k - 1) u_j(t, k) a^k = \sum_{k=0}^j (t^{j+1} - t^{k+1}) u_j(t, k) a^{k+1} = \sum_{k=1}^{j+1} (t^{j+1} - t^k) u_j(t, k-1) a^k,$$

where the last equality holds by changing the index. Hence, by comparing coefficients,

$(t^k - 1) u_j(t, k) = (t^{j+1} - t^k) u_j(t, k-1)$ for each $k = 1, 2, \dots, j$. Being true for some k ($2 \leq k \leq j$), we have the same equation true for $k-1$ and so $(t^{k-1} - 1) u_j(t, k-1) = (t^{j+1} - t^{k-1}) u_j(t, k-2)$ and thus

$$(t^{k-1} - 1)(t^k - 1) u_j(t, k) = (t^{j+1} - t^k)(t^{j+1} - t^{k-1}) u_j(t, k-2).$$

Continuing in this manner until we get to $u_j(t, 0) = 1$, we find

$$(t-1) \dots (t^{k-1} - 1)(t^k - 1) u_j(t, k) = (t^{j+1} - t^k)(t^{j+1} - t^{k-1}) \dots (t^{j+1} - t) u_j(t, 0)$$

or that

$$u_j(t, k) = \frac{(t^{j+1} - t)(t^{j+1} - t^2) \dots (t^{j+1} - t^k)}{(t-1)(t^2 - 1) \dots (t^k - 1)}.$$

By factoring out powers of t in the numerator and multiplying every factor in the numerator and denominator by -1 , we have

$$u_j(t, k) = \frac{t \cdot t^2 \cdot \dots \cdot t^k (1-t^j)(1-t^{j-1}) \dots (1-t^{j-k+1})}{(1-t)(1-t^2) \dots (1-t^k)}$$

$$u_j(t, k) = \begin{cases} t^{\binom{k+1}{2}} \frac{(1-t^j)(1-t^{j-1}) \dots (1-t^{j-k+1})}{(1-t)(1-t^2) \dots (1-t^k)} & \text{if } k < j \\ t^{\binom{j+1}{2}} & \text{if } k = j \end{cases}.$$

We shall use $[t^i]u_j(t, k)$ to denote the coefficient of t^i in $u_j(t, k)$. So, $[t^i]u_j(t, k)$ is the number of partitions of the integer i into k unequal parts having no part greater than j . So, for us with our interest in Cayley maps for dihedral groups, we, again, want $n-1$ unequal parts with no part greater than $2n-1$. So we are interested in $u_{2n-1}(t, n-1)$.

CHAPTER 2

EXISTING FORMULA

The goal of this thesis is to devise an improved formula for finding the average Cayley genus of Cayley maps with the dihedral group D_{2n} and generating set Δ consisting of all the reflections of D_{2n} . Prior to working on a new formula, we provide a study of the existing formula.

As we saw in Chapter 1, we can use Euler's formula (Theorem 1.1), to determine the genus of a particular Cayley map as long as we know the order of the graph, the size of the graph, and the number of regions in the embedding. The Cayley maps we are interested in are (D_{2n}, Δ, ρ) , where the group chosen is the dihedral group, which has order $4n$, and the generating set Δ consists of all the reflections. So, $\Delta = \{y, xy, x^2y, \dots, x^{2n-1}y\}$. Also, let ρ be a cyclic permutation of Δ , say $\rho = (y, x^{k_1}y, x^{k_2}y, \dots, x^{k_{2n-1}}y)$, where $\{k_1, k_2, \dots, k_{2n-1}\} = \{1, 2, \dots, 2n-1\}$. The voltage graph whose lift is the Cayley map (D_{2n}, Δ, ρ) is shown in Figure 2.A.

Recall that the Cayley graph is $G_\Delta(D_{2n}) = K_{2n, 2n}$. For this graph, the order p is the same as the order of the group, so that $p = 4n$. The size q can be found by noticing that the graph being a complete bipartite graph has each vertex in one of the partite sets (there are $2n$ here) adjacent to the $2n$ vertices in the other partite set. Hence, $q = 4n^2$.

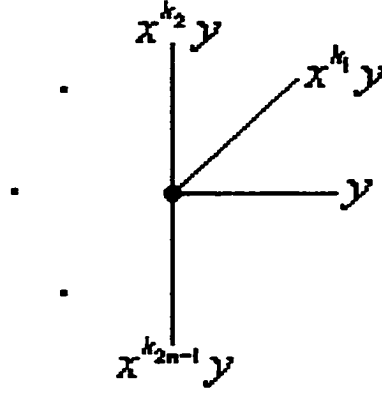


Figure 2.A Voltage Graph Embedding.

Using Theorem 1.4 we can find the number of the regions in the Cayley map (D_{2n}, Δ, ρ) . Since there is only one region in the voltage graph as seen in Figure 2.A, its lift will have $\frac{|\Gamma|}{s} = \frac{4n}{s}$ regions, where s is the order of the boundary element of the region that is being used. Thus, using $p = 4n$, $q = 4n^2$ and $r = \frac{4n}{s}$ in Euler's formula and solving for k we get:

$$p - q + r = 2 - 2k,$$

and by substituting,

$$4n - 4n^2 + \frac{4n}{s} = 2 - 2k,$$

and by rearranging,

$$2k = 2 - 4n + 4n^2 - \frac{4n}{s},$$

and solving for k ,

$$k = 1 + 2n \left[n - 1 - \frac{1}{s} \right].$$

So, to find the genus k given n it remains to find s . Again, s is the order of the boundary element. But the problem arising with finding the order for a boundary element is, we start out with $(2n-1)!$ possible cases of boundary elements taking into consideration all the possible cyclic arrangements of the generators. Before we find $(2n-1)!$ orders, we'd like to be able to reduce the number of cases if we can. We start by looking at exactly what a boundary element is. The general boundary element for the one region of our voltage graph is $yx^{k_1}yx^{k_2}yx^{k_3}\dots yx^{k_{2n-2}}yx^{k_{2n-1}}y$, where $\{k_1, k_2, \dots, k_{2n-1}\} = \{1, 2, \dots, 2n-1\}$. We group these elements in such a way to see that the boundary element is $(yx^{k_1})yx^{k_2}(yx^{k_3})\dots yx^{k_{2n-2}}(yx^{k_{2n-1}})y$. So group an x that has an exponent with an odd index of k with the y that comes before that x term. What this allows is to use the properties that $yx^{k_i} = x^{-k_i}y$ and $y^2 = 1$ to see that the boundary element is really just $x^{-k_1+k_2-k_3+k_4-\dots+k_{2n-2}-k_{2n-1}}$. So now we are trying to find $s = \text{ord}(x^{-k_1+k_2-k_3+k_4-\dots+k_{2n-2}-k_{2n-1}})$. We know that $\text{ord}(x) = 2n$ so that $\text{ord}(x^k) = \frac{2n}{\gcd(2n, k)}$.

What we do next is to reorder the exponent of the boundary element to find that we now need to find $\text{ord}(x^{k_2+k_4+k_6+\dots+k_{2n-2}-(k_1+k_3+k_5+\dots+k_{2n-1})})$. What this means is that we no longer have $(2n-1)!$ cases. All we need to do now is, in the new exponent, choose the $n-1$ values to be added from the $2n-1$ total values. The values that are left will be subtracted, and the order we add the values doesn't matter since addition is commutative.

So we now have $\binom{2n-1}{n-1}$ distinct choices for

$$\{k_2, k_4, k_6, \dots, k_{2n-2}\} \subseteq \{1, 2, 3, \dots, 2n-1\}.$$

In the $\binom{2n-1}{n-1}$ cases, it can happen that $A = \{k_2, k_4, \dots, k_{2n-2}\}$ and $A' = \{k_2', k_4', \dots, k_{2n-2}'\}$ with $A \neq A'$ but $\sum_{k_i \in A} k_i = \sum_{k_i' \in A'} k_i'$. So we can narrow down the cases further by finding all the different sums that are possible. Look at the possible combinations from the set $\{1, 2, 3, \dots, 2n-1\}$ that we can sum together. The least possible sum is $1+2+3+\dots+(n-2)+(n-1) = \binom{n}{2}$. The next sum that would be possible is given by $1+2+3+\dots+(n-2)+n = \binom{n}{2}+1$ and the sum after this is $1+2+3+\dots+(n-2)+(n+1) = \binom{n}{2}+2$ and so on until the final sum. The last sum is $(n+1)+(n+2)+\dots+(2n-1) = \binom{2n}{2} - \binom{n+1}{2} = 3\binom{n}{2}$. So all the possible sums are $\binom{n}{2}, \binom{n}{2}+1, \binom{n}{2}+2, \dots, 3\binom{n}{2}$. So this has reduced the number of cases to $2\binom{n}{2}+1$.

Also, we have a way to simplify the exponent in $x^{k_2+k_4+k_6+\dots+k_{2n-2}-(k_1+k_3+k_5+\dots+k_{2n-1})}$. In this exponent, let $S = k_2 + k_4 + k_6 + \dots + k_{2n-2}$. Then $k_1 + k_3 + k_5 + \dots + k_{2n-1} = \binom{2n}{2} - S$. So

$x^{k_2+k_4+k_6+\dots+k_{2n-2}-(k_1+k_3+k_5+\dots+k_{2n-1})} = x^{S - \left(\binom{2n}{2} - S\right)} = x^{2S - \binom{2n}{2}}$. So the cases at this point are for each

$S = \binom{n}{2}, \binom{n}{2}+1, \binom{n}{2}+2, \dots, 3\binom{n}{2}$, we must find the order of $x^{2S - \binom{2n}{2}}$.

Further, in order to be able to obtain our average Cayley genus, we must also count the number of different cyclic arrangements of generators that produce each such case.

Thus, we want to find how many different sets $A = \{k_2, k_4, \dots, k_{2n-2}\}$ give the same sum. Each set $A = \{k_2, k_4, \dots, k_{2n-2}\}$ gives a partition of $\sum_{k_i \in A} k_i$ into $n-1$ unequal parts, with no part greater than $2n-1$. A generating function for the number of partitions of an integer with k unequal parts, no part greater than j is given by $u_j(t, k)$ as seen in Chapter 1 and for our situation we are interested in $u_{2n-1}(t, n-1)$.

To reduce further, the same boundary element can occur for different sums. The following theorem states when this will occur.

Theorem 2.1 Let $A, B \subseteq \{1, 2, 3, \dots, 2n-1\}$ with $|A| = |B| = n-1$ and suppose

$$S_A = \sum_{a \in A} a \neq \sum_{b \in B} b = S_B. \text{ Then } x^{2S_A - \binom{2n}{2}} = x^{2S_B - \binom{2n}{2}} \text{ if and only if } S_A \equiv S_B \pmod{n}.$$

Proof. Observe that $x^{2S_A - \binom{2n}{2}} = x^{2S_B - \binom{2n}{2}}$ if and only if $2S_A - \binom{2n}{2} \equiv 2S_B - \binom{2n}{2} \pmod{2n}$ if and only if $2S_A \equiv 2S_B \pmod{2n}$ if and only if $S_A \equiv S_B \pmod{n}$. $\square$

Theorem 2.1 tells us that we now only need n cases, one for each $i = 0, 1, 2, \dots, n-1$. Now that the cases have been reduced, using the generating function $u_{2n-1}(t, n-1)$ we can state the existing formula for finding the average Cayley genus from Schultz [9].

Theorem 2.2 Let $n \geq 2$ be an integer and let $a = 0$ if n is even and $a = n$ if n is odd. Then the average Cayley genus $\bar{\gamma}(D_{2n}, \Delta)$, where Δ is the generating set for D_{2n} consisting of all the reflections, is given by

$$\bar{\gamma}(D_{2n}, \Delta) = \frac{n!(n-1)!}{(2n-1)!} \left\{ \sum_{i=0}^{n-1} \left[(2n^2 - 2n + 1 - \gcd(2n, a + 2i)) \times \sum_{j=0}^{n-1} \left[t^{\binom{i}{2} + i + jn} \right] u_{2n-1}(t, n-1) \right] \right\}$$

[Note: In Theorem 2.2, we take $\gcd(2n, 0)$ to mean $\gcd(2n, 2n)$ so that $\gcd(2n, 0) = \gcd(2n, 2n) = 2n$.]

Next, we will see this formula in use through an example. For the example, we will consider $n = 12$ and look at the table on the next page.

The numbers in the first column are all the values for i from 0 to $n-1$. The second column is $\gcd(2n, a + 2i)$ which is used in the formula. The third column is to find the corresponding genus, $2n^2 - 2n + 1 - \gcd(2n, a + 2i)$. The fourth column is using the generating function to find the coefficients needed for each i and then summing them. Then by using the values in the third and fourth columns in the table into the formula we get

$$\begin{aligned} \bar{\gamma}(D_{24}, \Delta) = & \frac{12!11!}{23!} [241(112720) + 263(112632) + 261(112707) + 259(112640) + \\ & 257(112710) + 263(112632) + 253(112716) + 263(112632) + 257(112710) + \\ & 259(112640) + 261(112707) + 263(112632)] = \frac{174631897}{869193}. \end{aligned}$$

Table 1 Example of $n = 12$ Using Existing Formula

i	gcd	genus	Sum of Generating Function Coefficients
0	24	241	$1+76+1109+6300+18320+30554+30554+18320+6300+1109+76+1 = 112720$
1	2	263	$1+98+1317+7040+19496+31132+29849+17125+5597+921+56 = 112632$
2	4	261	$2+129+1564+7850+20696+31641+29087+15968+4962+766+42 = 112707$
3	6	259	$3+165+1838+8698+21863+32017+28218+14812+4368+628+30 = 112640$
4	8	257	$5+212+2156+9613+23034+32312+27302+13703+3836+515+22 = 112710$
5	2	263	$7+266+2505+10560+24152+32467+26295+12608+3342+415+15 = 112632$
6	12	253	$11+336+2907+11573+25261+32540+25261+11573+2907+336+11 = 112716$
7	2	263	$15+415+3342+12608+26295+32467+24152+10560+2505+266+7 = 112632$
8	8	257	$22+515+3836+13703+27302+32312+23034+9613+2156+212+5 = 112710$
9	6	259	$30+628+4368+14812+28218+32017+21863+8698+1838+165+3 = 112640$
10	4	261	$42+766+4962+15968+29087+31641+20696+7850+1564+129+2 = 112707$
11	2	263	$56+921+5597+17125+29849+31132+19496+7040+1317+98+1 = 112632$

As seen in the table there are some problems to address with this formula, specifically, we make two observations. The first is that for different values of i , we have the same greatest common divisor in the second column, the same genus in the third column, and the same coefficient sums in the fourth column. What this suggests is that it may be possible to further reduce the number of cases necessary. Further, by narrowing down the cases, we would consequently arrive at a more accurate representation of the genus distribution. For now, we have the same genus occurring with repetition occasionally. We would like each one to appear only once. Secondly, in the fourth column, we have to go through a lot of tedious work using the generating function to find all the coefficients and then we have to find certain sums of them. We would like another way to determine the sums without having to resort to the generating function.

CHAPTER 3

PREPARATION FOR NEW FORMULA

We begin this chapter by recalling the generating function $u_{2n-1}(t, n-1)$. As seen in the previous chapter, this generating function appears in the existing formula for the average Cayley genus for the Cayley maps we are interested in. From Table 1, the example of $n=12$ using the existing formula, we see how the coefficients from the generating function are listed and then added together for each of the twelve cases. Additional examples of tables are given in Appendix A. These tables show the coefficients from the generating function $u_{2n-1}(t, n-1)$ for each $n=2, 3, \dots, 14$. The column at the right contains the sums of the coefficients in that row. As seen from the example and in the formula, it is these sums in this last column that we are interested in. The rows represent the n cases that are required for the existing formula.

So the tables keep track of the coefficients from the generating function. In particular, for a given n , if $u_{2n-1}(t, n-1) = \sum_{k=0}^{n(n-1)} a_k t^{\binom{n}{2}+k}$, then $a_0, a_1, \dots, a_{n-1}$ go into the first column of coefficients, $a_n, a_{n+1}, \dots, a_{2n-1}$ go into the second column of coefficients, and so on until $a_{n(n-1)}=1$ goes into first entry of the last column of coefficients, exhausting the coefficients of $u_{2n-1}(t, n-1)$. Then the coefficients of each row are summed up accordingly in the final column of the table. We let $c_0, c_1, c_2, \dots, c_{n-1}$

denote coefficient these sums, where then c_i is the sum of the coefficients from the generating function in the i -th row. After reviewing the tables, it is easily noted that the sum in the first row, c_0 , is different from $c_1, c_2, \dots, c_{n-1}$. These other rows all have $n-1$ coefficients whereas the row c_0 has n coefficients. So hopefully we can relate the other row sums in terms of this first row.

The first coefficient of the first row counts the number of partitions of $\binom{n}{2}$ and the other coefficients in this row count the number of partitions of

$$\binom{n}{2}+n, \binom{n}{2}+2n, \dots, 3\binom{n}{2}.$$

The coefficients in the first column, starting with the second row, count the partitions of $\binom{n}{2}+1, \binom{n}{2}+2, \dots, \binom{n}{2}+n-1$. Continuing in this manner in the second column, the coefficient that counts the partitions of $\binom{n}{2}+n$ is the first row, and the rest of the coefficients in this second column count the partitions of

$$\binom{n}{2}+n+1, \binom{n}{2}+n+2, \dots, \binom{n}{2}+2n-1.$$

We construct the rest of the columns in this same manner. That is, to find the coefficients in the j th column for $j=0, 1, 2, \dots, n-2$, we use the coefficient in the first row that we have already obtained for that column. Namely, the coefficient in the first row, j th column is counting the partitions of $\binom{n}{2}+jn$. Using this entry from the first row, the coefficient in this i th row that is in the j th column can be represented by

$\binom{n}{2} + jn + i$ for $i = 0, 1, 2, \dots, n-1$. From these tables, we see the importance of this first row, and especially the first row sum c_0 . What we would like to do is to be able to express the other row sums in terms of c_0 .

Now consider again the example of $n=12$ and the tables in Appendix A. We would like to know the sum of the coefficients of each row, that is, the entries in the final column of each of the tables, without having to use the generating function to list out each of the individual coefficients in their respective rows. In doing so, we will express each c_i , with $i = 1, 2, \dots, n-1$, in terms of c_0 . Our first step will be to set up a system of $n-1$ equations in the variables $c_1, c_2, \dots, c_{n-1}$.

Define $f(t) = \frac{1}{t^{\binom{n}{2}}} u_{2n-1}(t, n-1)$. We can rewrite it so that $f(t) = \sum_{k=0}^{n(n-1)} a_k t^k$, where

each a_k is the entry from the table as defined previously. What we would like to have is a closed formula for

$$c_0 = a_0 + a_n + a_{2n} + \dots + a_{n(n-2)} + a_{n(n-1)}$$

and when $1 \leq i \leq n-1$, a formula for

$$c_i = a_i + a_{n+i} + a_{2n+i} + \dots + a_{n(n-2)+i}.$$

The $n-1$ equations will come from considering each expression for $f(t)$ near the non-trivial n th roots of unity.

Here we will take a minute to recall some properties from abstract algebra. For a positive integer n , the set of all of the n th roots of unity forms a multiplicative group that

is cyclic. An n th root of unity that generates this multiplicative group is called a primitive n th root of unity.

It is also helpful to recall a few things from complex analysis. First, we define a complex number $z = a + bi = re^{i\theta} = r \cos \theta + ir \sin \theta$. We use complex numbers for their solutions to equations involving roots of unity. The n th roots of unity are the solutions to the equation $z^n = 1$, so they are of the form $z = e^{2\pi i \left(\frac{\ell}{n}\right)} = \cos\left(\frac{2\pi\ell}{n}\right) + i \sin\left(\frac{2\pi\ell}{n}\right)$, where $\ell = 0, 1, 2, \dots, n-1$. We will use a simpler notation for these solutions by defining $e^{2\pi i \left(\frac{\ell}{n}\right)} = e\left(\frac{\ell}{n}\right)$. Before proceeding any further we mention a few remarks.

Fact 1 If $c \in \mathbb{Z}$, then $e(c) = 1$. This comes from using the definition of $e\left(\frac{\ell}{n}\right)$, specifically $e(c) = e^{2\pi i(c)} = 1$.

Fact 2 For $e(\alpha)$ and $e(\beta)$ being solutions to $z^n = 1$, we have $e(\alpha + \beta) = e(\alpha) \cdot e(\beta)$.

We arrive at this fact by again using the definition. We have that $e(\alpha + \beta) = e^{2\pi i(\alpha + \beta)} = e^{2\pi i\alpha + 2\pi i\beta} = e^{2\pi i\alpha} e^{2\pi i\beta} = e(\alpha) \cdot e(\beta)$.

Fact 3 Let ξ be a primitive n th root of unity, then $\xi^{n-1} + \xi^{n-2} + \dots + \xi = -1$. This comes from using that 1 cannot be a primitive root of unity since 1 does not generate the other roots. So since ξ is a primitive n th root of unity, we have $\xi^n = 1$. Thus $\xi^n - 1 = 0$ and by factoring we have $(\xi - 1)(\xi^{n-1} + \xi^{n-2} + \dots + \xi + 1) = 0$. Since $\xi \neq 1$, we get that $\xi - 1 \neq 0$ so that $\xi^{n-1} + \xi^{n-2} + \dots + \xi + 1 = 0$. Thus we obtain fact 3. What this fact means is that the sum of all the nontrivial n th roots of unity is -1 .

Previously, it was noted that $\frac{1}{t^{\binom{n}{2}}} u_{2n-1}(t, n-1) = \frac{(1-t^{2n-1})(1-t^{2n-2})\dots(1-t^{n+1})}{(1-t^{n-1})(1-t^{n-2})\dots(1-t)}$ and

$\sum_{k=0}^{n(n-1)} a_k t^k$ were considered equal. This was because we were using them as generating

functions. For our purposes now, let $f(t) = \frac{(1-t^{2n-1})(1-t^{2n-2})\dots(1-t^{n+1})}{(1-t^{n-1})(1-t^{n-2})\dots(1-t)}$ and

$g(t) = \sum_{k=0}^{n(n-1)} a_k t^k$. Then $g(t)$ is defined for all complex numbers and $f(t)$ is defined on

$\mathbb{C} - \mathcal{N}$, where $\mathcal{N} = \bigcup_{N=1}^{n-1} \left\{ e\left(\frac{k}{N}\right) : 0 \leq k \leq N-1 \right\}$, i.e. $\mathcal{N}$ is the set of the N th roots of unity

for $N=1, 2, \dots, n-1$. Observe that $f(t) = g(t)$ for all $t \in \mathbb{C} - \mathcal{N}$. We know $g(t)$ is a polynomial and so $g(t)$ is continuous on $\mathbb{C}$. Thus, $\lim_{t \rightarrow t_0} g(t) = g(t_0)$ for all $t_0 \in \mathbb{C}$. But

since $f(t) = g(t)$ for all $t \in \mathbb{C} - \mathcal{N}$ and $\mathcal{N}$ is a finite set, it follows that $g(t_0) =$

$\lim_{t \rightarrow t_0} g(t) = \lim_{t \rightarrow t_0} f(t)$. So, we use $t_0 = e\left(\frac{\ell}{n}\right)$ for $\ell = 1, 2, \dots, n-1$ to get $n-1$ equations.

Now, we evaluate $g(t)$ at each of the n th roots of unity $e\left(\frac{\ell}{n}\right)$ for $\ell = 1, 2, \dots, n-1$.

Substituting into $g(t) = \sum_{k=0}^{n(n-1)} a_k t^k$, we have that $g\left(e\left(\frac{\ell}{n}\right)\right) = \sum_{k=0}^{n(n-1)} a_k \left(e\left(\frac{\ell}{n}\right)\right)^k$. By using

the identity that $e\left(\frac{\ell}{n}\right)^{r+ns} = e\left(\frac{\ell}{n}\right)^r$ we get

$$\sum_{k=0}^{n(n-1)} a_k \left(e\left(\frac{\ell}{n}\right)\right)^k =$$

$$\sum_{j=0}^{n-1} a_{nj} \left(e \left(\frac{\ell}{n} \right) \right)^{nj} + \sum_{j=0}^{n-2} a_{1+nj} \left(e \left(\frac{\ell}{n} \right) \right)^{1+nj} + \sum_{j=0}^{n-2} a_{2+nj} \left(e \left(\frac{\ell}{n} \right) \right)^{2+nj} + \dots + \sum_{j=0}^{n-2} a_{n-1+nj} \left(e \left(\frac{\ell}{n} \right) \right)^{n-1+nj} =$$

$$\sum_{j=0}^{n-1} a_{nj} + e \left(\frac{\ell}{n} \right) \sum_{j=0}^{n-2} a_{1+nj} + \left(e \left(\frac{\ell}{n} \right) \right)^2 \sum_{j=0}^{n-2} a_{2+nj} + \dots + \left(e \left(\frac{\ell}{n} \right) \right)^{n-1} \sum_{j=0}^{n-2} a_{n-1+nj}.$$

Now by definition, $c_0 = \sum_{j=0}^{n-1} a_{nj}$ and $c_i = \sum_{j=0}^{n-2} a_{i+nj}$ for each $i=1, 2, \dots, n-1$, so we get that

$$g \left(e \left(\frac{\ell}{n} \right) \right) = \sum_{k=0}^{n(n-1)} a_k \left(e \left(\frac{\ell}{n} \right) \right)^k = c_0 + c_1 e \left(\frac{\ell}{n} \right) + c_2 e \left(\frac{2\ell}{n} \right) + c_3 e \left(\frac{3\ell}{n} \right) + \dots + c_{n-1} e \left(\frac{(n-1)\ell}{n} \right)$$

for each $\ell=1, 2, \dots, n-1$. So we have $n-1$ equations in $n-1$ variables. Namely, the variables are $c_1, c_2, \dots, c_{n-1}$. So we can solve this system of equations by setting up a matrix equation. The matrix equation we get is given by the following:

$$\begin{bmatrix} e \left(\frac{1}{n} \right) & e \left(\frac{2}{n} \right) & \dots & e \left(\frac{n-1}{n} \right) \\ e \left(\frac{2}{n} \right) & e \left(\frac{4}{n} \right) & \dots & e \left(\frac{2(n-1)}{n} \right) \\ \vdots & \vdots & \dots & \vdots \\ e \left(\frac{n-1}{n} \right) & e \left(\frac{2(n-1)}{n} \right) & \dots & e \left(\frac{(n-1)(n-1)}{n} \right) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_{n-1} \end{bmatrix} = \begin{bmatrix} \lim_{t \rightarrow e \left(\frac{1}{n} \right)} f(t) - c_0 \\ \lim_{t \rightarrow e \left(\frac{2}{n} \right)} f(t) - c_0 \\ \vdots \\ \lim_{t \rightarrow e \left(\frac{n-1}{n} \right)} f(t) - c_0 \end{bmatrix}. \quad (*)$$

A comment here is needed about this matrix equation. The coefficient matrix satisfies the conditions for the Vandermonde matrix. This means that the determinant is non-zero, which implies that this equation not only has a solution, but that the solution is unique. We proceed to determine this solution. First, we calculate $\lim_{t \rightarrow e \left(\frac{\ell}{n} \right)} f(t)$.

Theorem 3.1 Let ℓ be an integer such that $1 \leq \ell \leq n-1$ and let $d = \gcd(n, \ell)$. Then

$$\lim_{t \rightarrow e\left(\frac{\ell}{n}\right)} f(t) = \binom{2d-1}{d-1}.$$

Proof. We will consider two cases.

Case 1. Suppose that $d=1$. In this case, $e\left(\frac{\ell}{n}\right)$ is a primitive n th root of unity so that if

$\xi = e\left(\frac{\ell}{n}\right)$, then $\xi^n = 1$. Since, $f(t)$ is defined at ξ and is continuous there, we have

$$\lim_{t \rightarrow \xi} f(t) = f(\xi) = \frac{(1-\xi^{2n-1})(1-\xi^{2n-2})\dots(1-\xi^{n+1})}{(1-\xi^{n-1})(1-\xi^{n-2})\dots(1-\xi)} = \frac{(1-\xi^{n-1}\xi^n)(1-\xi^{n-2}\xi^n)\dots(1-\xi\xi^n)}{(1-\xi^{n-1})(1-\xi^{n-2})\dots(1-\xi)} = 1.$$

Also when $d=1$, we have $\binom{2d-1}{d-1} = \binom{1}{0} = 1$.

Case 2. Suppose that d satisfies $1 < d < n$. Since $d = \gcd(n, \ell)$, we may write $n = dN$ and $\ell = dL$ for some integers N and L with $\gcd(N, L) = 1$. So $\lim_{t \rightarrow e\left(\frac{\ell}{n}\right)} f(t) =$

$\lim_{t \rightarrow e\left(\frac{dL}{dN}\right)} f(t) = \lim_{t \rightarrow e\left(\frac{L}{N}\right)} f(t)$. Since $\gcd(N, L) = 1$, it follows that $e\left(\frac{L}{N}\right)$ is a primitive N th

root of unity. So let $\xi = e\left(\frac{L}{N}\right)$ and then $\xi^N = 1$. Evaluating for $\lim_{t \rightarrow \xi} f(t)$, we obtain

$\lim_{t \rightarrow \xi} f(t) = \lim_{t \rightarrow \xi} \frac{(1-t^{2n-1})(1-t^{2n-2})\dots(1-t^{n+1})}{(1-t^{n-1})(1-t^{n-2})\dots(1-t)}$. Next, look at the general term

$\frac{(1-t^{n+i})}{(1-t^i)}$, for some i with $1 \leq i \leq n-1$, in this equation. Observe that if $N \nmid i$, then since

$n = dN$, $\xi^N = 1$, and $\xi^i \neq 1$ we obtain $\frac{(1-\xi^{n+i})}{(1-\xi^i)} = \frac{(1-\xi^i \xi^{dN})}{(1-\xi^i)} = 1$. Using this, we have

$$\lim_{t \rightarrow \xi} f(t) =$$

$$\begin{aligned} & \lim_{t \rightarrow \xi} \left(\prod_{\substack{1 \leq i \leq n-1 \\ N \nmid i}} \frac{1-t^{n+i}}{1-t^i} \right) \left(\prod_{\substack{1 \leq i \leq n-1 \\ N \nmid i}} \frac{1-t^{n+i}}{1-t^i} \right) = \lim_{t \rightarrow \xi} \left(\prod_{\substack{1 \leq i \leq n-1 \\ N \nmid i}} \frac{1-t^{n+i}}{1-t^i} \right) = \\ & \lim_{t \rightarrow \xi} \left(\frac{(1-t^{(2d-1)N}) \dots (1-t^{(d+2)N}) (1-t^{(d+1)N})}{(1-t^{(d-1)N}) \dots (1-t^{2N}) (1-t^N)} \right) = \lim_{t \rightarrow \xi} \left(\prod_{j=1}^{d-1} \frac{1-t^{(d+j)N}}{1-t^{jN}} \right) = \\ & \lim_{t \rightarrow \xi} \left(\prod_{j=1}^{d-1} \frac{(1-t^N)(1+t^N+t^{2N}+\dots+t^{(d+j-1)N})}{(1-t^N)(1+t^N+t^{2N}+\dots+t^{(j-1)N})} \right) = \\ & \lim_{t \rightarrow \xi} \left(\frac{1+t^N+t^{2N}+\dots+t^{(d+j-1)N}}{1+t^N+t^{2N}+\dots+t^{(j-1)N}} \right) = \prod_{j=1}^{d-1} \frac{d+j}{j} = \\ & \frac{(d+1)(d+2)\dots(2d-1)}{1 \cdot 2 \cdot 3 \cdot \dots \cdot (d-1)} = \frac{(2d-1)!}{d!(d-1)!} = \binom{2d-1}{d-1}. \quad \square \end{aligned}$$

Corollary 3.2 $\lim_{t \rightarrow e^{\left(\frac{\ell_1}{n}\right)}} f(t) = \lim_{t \rightarrow e^{\left(\frac{\ell_2}{n}\right)}} f(t)$ if and only if $\gcd(n, \ell_1) = \gcd(n, \ell_2)$.

Proof. Let $d = \gcd(n, \ell_1)$ and let $d' = \gcd(n, \ell_2)$. Then $\lim_{t \rightarrow e^{\left(\frac{\ell_1}{n}\right)}} f(t) = \binom{2d-1}{d-1}$ and

$\lim_{t \rightarrow e^{\left(\frac{\ell_2}{n}\right)}} f(t) = \binom{2d'-1}{d'-1}$. Thus, $\lim_{t \rightarrow e^{\left(\frac{\ell_1}{n}\right)}} f(t) = \lim_{t \rightarrow e^{\left(\frac{\ell_2}{n}\right)}} f(t)$ if and only if $d = d'$, that is, if

and only if $\gcd(n, \ell_1) = \gcd(n, \ell_2)$. $\square$

In order to solve the matrix equation (*), it will be useful to have the following notation for certain column vectors. Also, if $\bar{x}$ is a column vector, then we write $\bar{x}^T$ for the transpose of $\bar{x}$, meaning $\bar{x}^T$ is $\bar{x}$ written as a row vector. Let n be a positive integer and let d be a divisor of n with $n = dN$. Then define the $(n-1) \times 1$ column vectors as follows using the transpose notation:

$$\bar{c} = (c_1, c_2, \dots, c_{n-1})^T,$$

$$\bar{u} = (1, 1, \dots, 1)^T,$$

$$\bar{v}_d = (v_1, v_2, \dots, v_{n-1})^T \text{ where } v_k = \begin{cases} 1 & \text{if } d = \gcd(k, n) \\ 0 & \text{if } d \neq \gcd(k, n) \end{cases},$$

$$\bar{u}_d = (u_1, u_2, \dots, u_{n-1})^T \text{ where } u_k = \begin{cases} 0 & \text{if } N \mid k \\ 1 & \text{if } N \nmid k \end{cases}.$$

Then observe that

$$\bar{u} - \bar{u}_d = (1 - u_1, 1 - u_2, \dots, 1 - u_{n-1})^T \text{ where } 1 - u_k = \begin{cases} 1 & \text{if } N \mid k \\ 0 & \text{if } N \nmid k \end{cases}.$$

Also let A be the coefficient matrix of the equation (*) and $\bar{b}$ be the solution vector of (*). Thus observe that if $D = \{d : d \mid n, 1 < d < n\}$, then

$$\bar{b} = (1 - c_0)\bar{u} + \sum_{d \in D} \left(\binom{2d-1}{d-1} - 1 \right) \bar{v}_d.$$

Using these vectors, the equation (*) is $A\bar{c} = \bar{b}$. We proceed to present several lemmas that will be useful in solving this equation.

Lemma 3.3 Using A and $\bar{u}$ as defined previously, $A\bar{u} = (-1, -1, \dots, -1)^T$.

Proof. Let $A\bar{u} = (k_1, k_2, \dots, k_{n-1})^T$ so that k_j is the sum of the j th row of the matrix A , that is, $k_j = \sum_{\ell=1}^{n-1} e\left(\frac{j\ell}{n}\right)$ for each j where $1 \leq j \leq n-1$. If $\gcd(j, n) = 1$, then by Fact 3 we have $\sum_{\ell=1}^{n-1} e\left(\frac{j\ell}{n}\right) = -1$. So suppose that the $\gcd(j, n) = d$, where $d \neq 1$. Then we may write $j = Jd$ and $n = Nd$ for integers J and N with $\gcd(J, N) = 1$. By grouping terms as follows,

$$\begin{aligned} \sum_{\ell=1}^{n-1} e\left(\frac{j\ell}{n}\right) &= \sum_{\ell=1}^{n-1} e\left(\frac{dJ\ell}{dN}\right) = \sum_{\ell=1}^{n-1} e\left(\frac{J\ell}{N}\right) = \left[e\left(\frac{J}{N}\right) + e\left(\frac{2J}{N}\right) + \dots + e\left(\frac{(N-1)J}{N}\right) + e\left(\frac{NJ}{N}\right) \right] + \\ &\left[e\left(\frac{(N+1)J}{N}\right) + e\left(\frac{(N+2)J}{N}\right) + \dots + e\left(\frac{(2N-1)J}{N}\right) + e\left(\frac{2NJ}{N}\right) \right] + \dots + \left[e\left(\frac{(d-1)NJ}{N}\right) + e\left(\frac{[(d-1)N+1]J}{N}\right) + \right. \\ &\left. e\left(\frac{[(d-1)N+2]J}{N}\right) + \dots + e\left(\frac{(dN-1)J}{N}\right) \right], \end{aligned}$$

we see that we have added all N N th roots of unity $d-1$ times and then have added the $N-1$ nontrivial N th roots of unity once. Thus, using Fact 3, we obtain

$$\sum_{\ell=1}^{n-1} e\left(\frac{j\ell}{n}\right) = (d-1) \left[\sum_{\ell=1}^N e\left(\frac{J\ell}{N}\right) \right] + \sum_{\ell=1}^{N-1} e\left(\frac{J\ell}{N}\right) = (d-1) \cdot (0) + (-1) = -1. \quad \square$$

So Lemma 3.3 gives us that $A[(c_0 - 1)\bar{u}] = (1 - c_0)\bar{u}$, which is the first term in our solution vector $\bar{b}$. Later we define values U_N for each $N \in D$ such that

$A \left(\sum_{N \in D} U_N \bar{u}_N \right) = \sum_{d \in D} \left(\binom{2d-1}{d-1} - 1 \right) \bar{v}_d$, the second part of our solution vector. However, it

is useful to first provide a few more helpful observations.

Lemma 3.4 Let $N \in D$ and $n = dN$. Then $A\bar{u}_N = -N(\bar{u} - \bar{u}_d)$.

Proof. Notice that $\bar{u}_N = (u_1, u_2, \dots, u_{n-1})^T$, where $u_k = \begin{cases} 0 & \text{if } d \mid k \\ 1 & \text{if } d \nmid k \end{cases}$. Then

$A\bar{u}_N = (i_1, i_2, \dots, i_{n-1})$, where $i_k = \sum_{\substack{1 \leq \ell \leq n-1 \\ d \nmid \ell}} e\left(\frac{k\ell}{n}\right)$. Since the sum of the entries in an entire

row of A is -1 , we have that $\sum_{\substack{1 \leq \ell \leq n-1 \\ d \nmid \ell}} e\left(\frac{k\ell}{n}\right) = -1 - \sum_{\substack{1 \leq \ell \leq n-1 \\ d \mid \ell}} e\left(\frac{k\ell}{n}\right) = -1 - \sum_{\substack{1 \leq \ell \leq n-1 \\ \ell = dL}} e\left(\frac{k\ell}{n}\right) =$

$-1 - \sum_{L=1}^{N-1} e\left(\frac{kL}{N}\right)$. If $N \mid k$, then $k = NK$ for some integer K so that by using fact 1 $A\bar{u}_N =$

$-1 - \sum_{L=1}^{N-1} e\left(\frac{NKL}{N}\right) = -1 - \sum_{L=1}^{N-1} e(KL) = -1 - (N-1) = -N$. If $N \nmid k$, then $k = NK + r$ for

some integers K and r with $1 \leq r \leq N-1$ so that using facts 1 and 3, we have $A\bar{u}_N =$

$$-1 - \sum_{L=1}^{N-1} e\left(\frac{(NK+r)L}{N}\right) = -1 - \sum_{L=1}^{N-1} e\left(\frac{NKL}{N}\right) e\left(\frac{rL}{N}\right) = -1 - \sum_{L=1}^{N-1} e\left(\frac{rL}{N}\right) = -1 - (-1) = 0.$$

So $A\bar{u}_N = \begin{cases} -N & \text{if } N \mid k \\ 0 & \text{if } N \nmid k \end{cases}$. Thus, since $\bar{u} - \bar{u}_d = (1 - u_1, 1 - u_2, \dots, 1 - u_{n-1})^T$, where

$1 - u_k = \begin{cases} 1 & \text{if } N \mid k \\ 0 & \text{if } N \nmid k \end{cases}$, it follows that $A\bar{u}_N = -N(\bar{u} - \bar{u}_d)$. $\square$

Lemma 3.5 Let $D = \{d : d \mid n, 1 < d < n\}$ and let $N \in D$. Then $\bar{u} - \bar{u}_{n/N} = \sum_{\substack{d \in D \\ N \nmid d}} \bar{v}_d$.

Proof. Previously, it was noted that $\bar{u} - \bar{u}_{n/N} = (i_1, i_2, \dots, i_{n-1})^T$ where $i_k = \begin{cases} 1 & \text{if } N \mid k \\ 0 & \text{if } N \nmid k \end{cases}$. Let $\mathcal{X} = \{d \in D : N \mid d\} = \{\ell_1, \ell_2, \dots, \ell_a\}$. So, $\mathcal{X}$ is the set of positive proper divisors of n that are also multiples of N . (For example, when $n=12$ and $N=2$ we have $D = \{2, 3, 4, 6\}$ and $\mathcal{X} = \{2, 4, 6\}$.) So $\sum_{\substack{d \in D \\ N \nmid d}} \bar{v}_d = \sum_{\mathcal{X}} \bar{v}_{\ell} = \bar{v}_{\ell_1} + \bar{v}_{\ell_2} + \dots + \bar{v}_{\ell_a} = (v_1, v_2, \dots, v_{n-1})^T$ where, for each $k=1, 2, \dots, n-1$, we have $v_k = \begin{cases} 1 & \text{if } \gcd(k, n) \in \mathcal{X} \\ 0 & \text{if } \gcd(k, n) \notin \mathcal{X} \end{cases}$. We verify that $v_k = i_k$ for each $k=1, 2, \dots, n-1$. If $\gcd(k, n) \in \mathcal{X}$, then $k = \ell_j K$ for some $\ell_j \in \mathcal{X}$ and some integer K . Since $N \mid \ell_j$, we have $\ell_j = NL_j$ for some integer L_j , so that $k = NL_j K$. So $N \mid k$ in this case. On the other hand, if $\gcd(k, n) = g \notin \mathcal{X}$, then $N \nmid g$ by the defining property of $\mathcal{X}$. Since $N \mid n$, we know $N \nmid k$ for otherwise we would have $N \mid g$, a contradiction. Thus, $v_k = i_k$ for each $k=1, 2, \dots, n-1$. $\square$

What these two lemmas allow us to do is to show the relationship between the $\bar{u}_N$ vectors, where $N \in D$, as a sum of vectors $\bar{v}_d$, for $d \in D$ as well. This will prove useful in finding the second part of the solution vector $\bar{b}$.

Now we are ready to define the values U_N for $N \in D$ such that

$$A\left(\sum_{N \in D} U_N \bar{u}_N\right) = \sum_{d \in D} \left(\binom{2d-1}{d-1} - 1 \right) \bar{v}_d.$$

We define

$$U_N = \frac{\binom{2N-1}{N-1} - 1 - \sum_{\substack{1 < k < N \\ k|N}} k U_k}{N}.$$

Looking into this definition, we see that it is recursive. What this means is that to find a particular U_N , it depends on what values U_k we get for the divisors k of N with $1 < k < N$, if N has any such divisors.

Recall the solution vector $\bar{b}$ and the expression $\left(\binom{2d-1}{d-1} - 1 \right) \bar{v}_d$ that lies within it.

This leads to the following lemma in which we will use the definition of U_N .

Lemma 3.6 For each $d \in D$, the sum $\sum_{\substack{N \in D \\ N|d}} NU_N = \binom{2d-1}{d-1} - 1$.

Proof. Observe $\sum_{\substack{N \in D \\ N|d}} NU_N = \sum_{\substack{1 < N < d \\ N|d}} NU_N + dU_d = \sum_{\substack{1 < N < d \\ N|d}} NU_N + \left(\binom{2d-1}{d-1} - 1 \right) - \sum_{\substack{1 < N < d \\ N|d}} NU_N =$
 $\left(\binom{2d-1}{d-1} - 1 \right).$ $\square$

With the help of the previous four lemmas, we are now prepared to solve equation $(*)$, in other words, we are now ready to solve $A\bar{c} = \bar{b}$ for $\bar{c}$. We do this in the following theorem.

Theorem 3.7 The solution to the matrix equation $A\bar{c} = \bar{b}$ shown as (*) is

$$\bar{c} = (c_0 - 1)\bar{u} - \sum_{N \in D} U_N \bar{u}_N$$

Proof. Consider $A\bar{c} = A \left[(c_0 - 1)\bar{u} - \sum_{N \in D} U_N \bar{u}_N \right]$.

By Lemma 3.3, we have

$$A\bar{c} = (1 - c_0)\bar{u} - \sum_{N \in D} AU_N \bar{u}_N.$$

Using Lemma 3.4, it follows that

$$A\bar{c} = (1 - c_0)\bar{u} + \sum_{N \in D} NU_N (\bar{u} - \bar{u}_{n/N}).$$

Applying Lemma 3.5, we obtain

$$A\bar{c} = (1 - c_0)\bar{u} + \sum_{N \in D} NU_N \left(\sum_{\substack{d \in D \\ N|d}} \bar{v}_d \right) = (1 - c_0)\bar{u} + \sum_{d \in D} \left(\sum_{\substack{N \in D \\ N|d}} NU_N \right) \bar{v}_d.$$

Finally, by Lemma 3.6,

$$A\bar{c} = (1 - c_0)\bar{u} + \sum_{d \in D} \left(\left(\binom{2d-1}{d-1} - 1 \right) \bar{v}_d \right) = \bar{b}. \quad \square$$

Before going further, we shall introduce the Euler ϕ -function $\phi(m)$, which denotes the number of positive integers less than m that are relatively prime to m and $\phi(1)$ is defined to be 1. Then, for each c_i where $i \in D$, the number of elements in $\{c_i : 1 \leq i \leq n-1, \gcd(i, n) = d\}$ for a given d is $\phi\left(\frac{n}{d}\right)$. Thus narrowing down the number of cases even further. Instead of n cases, we now only need one case for each

divisor d of n where $1 \leq d < n$ as well as the initial case of c_0 . A basic result of number theory (see [7], for example) that will be useful in the following proof is that

$$\sum_{d|n} \phi(d) = \sum_{d|n} \phi\left(\frac{n}{d}\right) = n. \quad \text{For our purposes, we will use that } \sum_{\substack{1 \leq d < n \\ d|n}} \phi\left(\frac{n}{d}\right) = n-1. \quad \text{This}$$

follows directly since $\sum_{\substack{1 \leq d < n \\ d|n}} \phi\left(\frac{n}{d}\right) = \left[\sum_{d|n} \phi\left(\frac{n}{d}\right) \right] - \phi\left(\frac{n}{n}\right) = n-1$. We now use this in the

following corollary of Theorem 3.7. Also, we use the notation $D^* = D \cup \{1\}$.

Corollary 3.8 Let $n \geq 2$ be a positive integer. Let $u_{2n-1}(t, n-1) = \sum_{k=0}^{n(n-1)} a_k t^{\binom{n}{2}+k}$ be the

generating function for partitions of integers with $n-1$ unequal parts and no part greater

than $2n-1$. Define $c_0 = \sum_{j=0}^{n-1} a_{nj}$ and for each $i = 1, 2, \dots, n-1$, let $c_i = \sum_{j=0}^{n-2} a_{i+nj}$. Then

$c_i = c_j$ if and only if $\gcd(i, n) = \gcd(j, n)$. Furthermore,

$$c_0 = \frac{\binom{2n-1}{n-1} + n-1 + \sum_{k \in D^*} \phi\left(\frac{n}{k}\right) \sum_{\substack{N \in D \\ n \nmid kN}} U_N}{n}$$

and for each $i \in D^*$, the value

$$c_i = c_0 - \left[1 + \sum_{\substack{N \in D \\ n \nmid iN}} U_N \right].$$

Proof. That $c_i = c_j$ if and only if $\gcd(i, n) = \gcd(j, n)$ follows from Corollary 3.2.

For each $i \in D^*$, the value $c_i = c_0 - \left[1 + \sum_{\substack{N \in D \\ n \nmid iN}} U_N \right]$ follows from Theorem 3.7. Finally, to

solve for c_0 , we know that the total $c_0 + \sum_{i=1}^{n-1} c_i = \binom{2n-1}{n-1}$ so that $\sum_{i=1}^{n-1} c_i = \binom{2n-1}{n-1} - c_0$. By

the comments preceding this corollary, we have that

$$\begin{aligned} \binom{2n-1}{n-1} - c_0 &= \sum_{i=1}^{n-1} c_i = \sum_{k \in D^*} \phi\left(\frac{n}{k}\right) c_k = \\ &= \sum_{k \in D^*} \left[\phi\left(\frac{n}{k}\right) \left[(c_0 - 1) - \sum_{\substack{N \in D \\ n \nmid kN}} U_N \right] \right] = \\ &= \sum_{k \in D^*} \left[\phi\left(\frac{n}{k}\right) (c_0 - 1) \right] - \sum_{k \in D^*} \left[\phi\left(\frac{n}{k}\right) \sum_{\substack{N \in D \\ n \nmid kN}} U_N \right] = \\ &= (c_0 - 1)(n - 1) - \sum_{k \in D^*} \left[\phi\left(\frac{n}{k}\right) \sum_{\substack{N \in D \\ n \nmid kN}} U_N \right]. \end{aligned}$$

Hence,

$$\binom{2n-1}{n-1} - c_0 = (c_0 - 1)(n - 1) - \sum_{k \in D^*} \left[\phi\left(\frac{n}{k}\right) \sum_{\substack{N \in D \\ n \nmid kN}} U_N \right]$$

and solving for c_0 , we have

$$c_0 = \frac{\binom{2n-1}{n-1} + n - 1 + \sum_{k \in D^*} \left[\phi\left(\frac{n}{k}\right) \sum_{\substack{N \in D \\ n \nmid kN}} U_N \right]}{n}. \quad \square$$

So now, we have reduced the number of cases from n to the number of divisors of n . To show what an improvement this is, from Hardy [6], we have that the number of divisors of n is $d(n) = O(n^\delta)$ for all positive δ . Also, from Corollary 3.8, we are now able to determine the coefficient sums that we have been looking for without having to use the generating function. These were our two main goals in achieving a new, simplified formula for finding the average Cayley genus for dihedral groups.

CHAPTER 4

NEW FORMULA AND SPECIAL CASES

In this chapter, we will devise a new formula for finding the average Cayley genus for dihedral groups. Recall that we are working with Cayley maps for D_{2n} with $\Delta = \{y, xy, x^2y, \dots, x^{2^{n-1}}y\}$. A particular Cayley map (D_{2n}, Δ, ρ) is the lift of a voltage graph that has one region. That region has an associated boundary element and the genus of the Cayley map in question is $k = 1 + 2n \left[n - 1 - \frac{1}{s} \right]$, where s is the order of this boundary element of the region from the voltage graph embedding. From our work in Chapter 3, we need only find the order of the boundary elements that correspond to the terms c_0 and c_d , where $d \mid n$ and $1 \leq d < n$. We consider these in two cases.

We first consider c_0 and try to find the order of the boundary element of the region corresponding to c_0 . From Chapter 2 (page 17), recall that this boundary element is

$$x^{2\binom{n}{2} - \binom{2n}{2}}. \text{ Simplifying the exponent, we obtain } 2\binom{n}{2} - \binom{2n}{2} = \frac{2 \cdot n!}{2!(n-2)!} - \frac{(2n)!}{2!(2n-2)!} =$$

$$\frac{2n(n-1)}{2} - \frac{2n(2n-1)}{2} = n(n-1) - n(2n-1) = n[(n-1) - (2n-1)] = n[-n] = -n^2. \text{ Recall}$$

that the order of x is $2n$, so that the exponent $2\binom{n}{2} - \binom{2n}{2}$ is $-n^2 \equiv n^2 \pmod{2n}$. Let

$a \equiv n^2 \pmod{2n}$ with $a \in \{0, 1, 2, \dots, 2n-1\}$. Then $a = \begin{cases} 0 & \text{if } n \text{ is even} \\ n & \text{if } n \text{ is odd} \end{cases}$. Now that we

know what the boundary element is, we can find the order of it. The $\text{ord}(x^a) =$

$$\frac{2n}{\gcd(2n, a)} = \begin{cases} 1 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd} \end{cases}. \quad (\text{Recall that we use } \gcd(2n, 0) = \gcd(2n, 2n) = 2n.)$$

Now, we consider c_d and find its boundary element and the order of that boundary element. Again, from page 17, we have that the boundary element is $x^{2\binom{n}{2}+2d-\binom{2n}{2}}$.

Through similar calculations, we get that the exponent $2\binom{n}{2}+2d-\binom{2n}{2} = -n^2 + 2d$. So

now let $b \in \{0, 1, 2, \dots, 2n-1\}$ such that

$$b \equiv (-n^2 + 2d) \pmod{2n}.$$

Then $b = \begin{cases} 2d & \text{if } n \text{ is even} \\ n-2d & \text{if } n \text{ is odd} \end{cases}$. Now that we have the boundary elements for c_d , we can

find the order of these elements. The $\text{ord}(x^b) = \frac{2n}{\gcd(2n, b)} = \begin{cases} \frac{n}{d} & \text{if } n \text{ is even} \\ \frac{2n}{d} & \text{if } n \text{ is odd} \end{cases}$. Now

that we have the orders of the boundary elements, we can find the corresponding genera.

Recall, from the beginning of Chapter 2, the genus was calculated to be

$k = 1 + 2n \left[n - 1 - \frac{1}{s} \right]$. So the genus depends on the order of the boundary element, and

this depends on which case, c_0 or c_d , we have and whether n is even or odd, as seen in finding the orders for the boundary elements.

First, we will find the genus corresponding to c_0 . The order of the boundary element

for the case c_0 is given by $\text{ord}(x^a) = \begin{cases} 1 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd} \end{cases}$. This gives us that the genus k is

given by $k = \begin{cases} 2n^2 - 4n + 1 & \text{if } n \text{ is even} \\ 2n^2 - 3n + 1 & \text{if } n \text{ is odd} \end{cases}$. Similarly, we now find the genus corresponding

to c_d . The order of the boundary element for the case c_d , where $d|n$, from above, is

given by $\text{ord}(x^b) = \begin{cases} \frac{n}{d} & \text{if } n \text{ is even} \\ \frac{2n}{d} & \text{if } n \text{ is odd} \end{cases}$. Thus, the genus k for this case is given by

$k = \begin{cases} 2n^2 - 2n - 2d + 1 & \text{if } n \text{ is even} \\ 2n^2 - 2n - d + 1 & \text{if } n \text{ is odd} \end{cases}$. Now that we have formulas for each of the Cayley

genera, we can now calculate the average Cayley genus.

Theorem 4.1 Let $n \geq 2$ be an integer. Let $D = \{d : d|n, 1 < d < n\}$ and

$D^* = D \cup \{1\}$. For $N \in D$ and $i \in D^*$, let

$$U_N = \frac{\binom{2N-1}{N-1} - 1 - \sum_{\substack{k|N \\ 1 < k < N}} k \cdot U_k}{N},$$

$$c_0 = \frac{\binom{2n-1}{n-1} + n - 1 + \sum_{k \in D^*} \phi\left(\frac{n}{k}\right) \sum_{\substack{N \in D \\ n|kN}} U_N}{n},$$

$$c_i = c_0 - \left[1 + \sum_{\substack{N \in D \\ n|iN}} U_N \right].$$

Then the average Cayley genus $\bar{\gamma}(D_{2n}, \Delta)$ where D_{2n} is the dihedral group and Δ is the generating set consisting of all the reflections is given by

$$\bar{\gamma}(D_{2n}, \Delta) = \begin{cases} \binom{2n-1}{n-1}^{-1} \left[c_0(2n^2 - 4n + 1) + \sum_{d \in D^*} c_d \phi\left(\frac{n}{d}\right) (2n^2 - 2n - 2d + 1) \right] & \text{if } n \text{ is even} \\ \binom{2n-1}{n-1}^{-1} \left[c_0(2n^2 - 3n + 1) + \sum_{d \in D^*} c_d \phi\left(\frac{n}{d}\right) (2n^2 - 2n - d + 1) \right] & \text{if } n \text{ is odd.} \end{cases}$$

What we will do now is to revisit the example of $n=12$. We shall make a new table using the new formula. However, we have to recognize that we have to work out the part with c_0 separately from the rest of the divisors $D^* = \{1, 2, 3, 4, 6\}$. With n being even, we use the first equation in the formula and build the following table.

Table 2 Example of $n=12$ Using New Formula.

d	$2n^2 - 4n + 1$	c_0
0	241	112720

d	$\phi\left(\frac{n}{d}\right)$	$2n^2 - 2n - 2d + 1$	c_d
1	4	263	112632
2	2	261	112707
3	2	259	112640
4	2	257	112710
6	1	253	112716

Now, we use the values from the table in the formula for when n is even. We obtain

$$\begin{aligned} \bar{\gamma}(D_{24}, \Delta) &= \frac{12!11!}{23!} \left[241(112720) + 4 \cdot 263(112632) + 2 \cdot 261(112707) + \right. \\ &\quad \left. 2 \cdot 259(112640) + 2 \cdot 257(112710) + 253(112716) \right] = \frac{174631897}{869193}. \end{aligned}$$

This being the same average Cayley genus that we obtained from Table 1 in Chapter 2 using the existing formula. Comparing Table 1 with Table 2 that we just constructed for finding the average Cayley genus when $n = 12$, we can see the two major problems in the existing formula and the reasons for the new formula. The new formula as seen in Table 2 reduced the number of cases from twelve to six. Also, as can be seen in the final column of both tables, using the new formula, we directly achieve the coefficient sums. So this new formula provides the improvements we were hoping for.

Now, to look at some special cases using this formula. The first special case we will look at is when $n = p$, where p is an odd prime. Being that it is an odd prime, the only divisor d of p that satisfies $1 \leq d < p$ is 1. So for an odd prime integer, there are only two cases, c_0 and c_1 . Using Corollary 3.8, we have that

$$c_0 = \frac{\binom{2p-1}{p-1} + p - 1}{p}$$

and

$$c_1 = c_0 - 1 = \frac{\binom{2p-1}{p-1} - 1}{p}.$$

And from Theorem 4.1 when n is odd, we obtain a formula that depends only on p .

Corollary 4.2 If p is an odd prime, then

$$\bar{\gamma}(D_{2p}, \Delta) = \binom{2p-1}{p-1}^{-1} \left[\left(\frac{\binom{2p-1}{p-1} + p - 1}{p} \right) (2p^2 - 3p + 1) + 2(p-1)^2 \left(\binom{2p-1}{p-1} - 1 \right) \right].$$

Another special case we will consider is when $n = p^2$ where again p is an odd prime.

So the only divisors of p^2 satisfying $1 \leq d < p^2$ are 1 and p . So this leads us to three cases, c_0 , c_1 and c_p . From Corollary 3.8,

$$c_0 = \frac{\binom{2p^2-1}{p^2-1} + p^2 - 1 + (p-1) \left(\binom{2p-1}{p-1} - 1 \right)}{p^2},$$

$$c_1 = c_0 - 1 - U_p = \frac{\binom{2p^2-1}{p^2-1} - \binom{2p-1}{p-1}}{p^2},$$

and

$$c_p = c_0 - 1 = \frac{\binom{2p^2-1}{p^2-1} - 1 + (p-1) \left(\binom{2p-1}{p-1} - 1 \right)}{p^2}.$$

Using Theorem 4.1 when n is odd, we again obtain a formula that depends only on p .

Corollary 4.3 If p is an odd prime, then

$$\begin{aligned} \bar{\gamma}(D_{2p^2}, \Delta) = & \binom{2p^2-1}{p^2-1}^{-1} \left[\left(\frac{\binom{2p^2-1}{p^2-1} + p^2 - 1 + (p-1) \left(\binom{2p-1}{p-1} - 1 \right)}{p^2} \right) (2p^4 - 3p^2 + 1) + \right. \\ & 2p(p^2-1)(p-1) \left(\binom{2p^2-1}{p^2-1} - \binom{2p-1}{p-1} \right) + \\ & \left. \left(\frac{\binom{2p^2-1}{p^2-1} - 1 + (p-1) \left(\binom{2p-1}{p-1} - 1 \right)}{p^2} \right) (p-1)(2p^4 - 2p^2 - p + 1) \right]. \end{aligned}$$

A third special case we will look at is when $n = 2p$, where again p is an odd prime.

So the divisors of $2p$ satisfying $1 \leq d < 2p$ are 1, 2 and p . Thus, we have four cases to look at, namely c_0 , c_1 , c_2 and c_p . Using Corollary 3.8 to find c_0 , c_1 , c_2 and c_p we get that

$$c_0 = \frac{\binom{4p-1}{2p-1} + \binom{2p-1}{p-1} + 4p - 4}{2p},$$

$$c_1 = c_0 - 1 - U_2 - U_p = \frac{\binom{4p-1}{2p-1} - \binom{2p-1}{p-1} - 2}{2p},$$

$$c_2 = c_0 - 1 - U_2 = \frac{\binom{4p-1}{2p-1} + \binom{2p-1}{p-1} - 4}{2p},$$

and

$$c_p = c_0 - 1 - U_p = \frac{\binom{4p-1}{2p-1} - \binom{2p-1}{p-1} + 2p - 2}{2p}.$$

From Theorem 4.1 when n is even, we obtain a formula that depends solely on p .

Corollary 4.4 If $n = 2p$, where p is an odd prime, then

$$\begin{aligned} \bar{\gamma}(D_{4p}, \Delta) = & \binom{4p-1}{2p-1}^{-1} \left[\left(\frac{\binom{4p-1}{2p-1} + \binom{2p-1}{p-1} + 4p-4}{2p} \right) (8p^2 - 8p + 1) + \right. \\ & (p-1) \left(\frac{\binom{4p-1}{2p-1} - \binom{2p-1}{p-1} - 2}{2p} \right) (8p^2 - 4p - 1) + \\ & (p-1) \left(\frac{\binom{4p-1}{2p-1} + \binom{2p-1}{p-1} - 4}{2p} \right) (8p^2 - 4p - 3) + \\ & \left. \left(\frac{\binom{4p-1}{2p-1} - \binom{2p-1}{p-1} + 2p - 2}{2p} \right) (8p^2 - 6p + 1) \right]. \end{aligned}$$

In conclusion, we now have developed a new formula for finding the average Cayley genus for the dihedral group with generating set consisting of all the reflections. This formula is an improvement in that it allows for fewer cases and enables us to find directly the coefficient sums without having to use the generating function. What sparked our interest in improving this formula can best be seen by looking at the tables in Appendix A. In particular, the prime case seemed the most interesting. For example, look at $n = 13$. What was intriguing the most is what happens in the rows $c_1, c_2, \dots, c_6$. From the generating function, we see six totally different sets of coefficients, but when summed

up in each row, each sums to the same answer. After noticing this pattern for the prime case, we tried to see if there were similar patterns in other cases. That's when we noticed the patterns in the p^2 and $2p$ cases. This inspired us to then try and simplify the formula more. Now this need not be the end for the average Cayley genus for Cayley maps involving the dihedral group. For example we only considered the generating set Δ consisting of all the reflections of the dihedral group. What would happen if we now looked at the generating set to be a subset of the reflections? Also, knowing that for other groups, finding the average Cayley genus has been difficult to find, can the techniques used in this paper be used for determining the average Cayley genus for Cayley maps using another group other than the dihedral group? These questions, as well as others still await further work.

APPENDIX A

GENERATING FUNCTION COEFFICIENTS FOR $n = 2, 3, \dots, 14$

n=2			
1	1	1	2
2	1		1

n=3				
3	1	2	1	4
4	1	2		3
5	2	1		3

n=4					
6	1	4	4	1	10
7	1	4	3		8
8	2	5	2		9
9	3	4	1		8

n=5						
10	1	6	12	6	1	26
11	1	8	11	5		25
12	2	9	11	3		25
13	3	11	9	2		25
14	5	11	8	1		25

n=6							
15	1	10	29	29	10	1	80
16	1	12	30	25	7		75
17	2	16	32	23	5		78
18	3	19	32	19	3		76
19	5	23	32	16	2		78
20	7	25	30	12	1		75

n=7								
21	1	14	61	94	61	14	1	246
22	1	19	68	94	52	11		245
23	2	24	76	90	46	7		245
24	3	31	81	88	37	5		245
25	5	37	88	81	31	3		245
26	7	46	90	76	24	2		245
27	11	52	94	68	19	1		245

n=8									
28	1	21	120	263	263	120	21	1	810
29	1	27	137	272	247	101	15		800
30	2	36	158	282	233	86	11		808
31	3	45	176	285	214	70	7		800
32	5	58	197	289	197	58	5		809
33	7	70	214	285	176	45	3		800
34	11	86	233	282	158	36	2		808
35	15	101	247	272	137	27	1		800

n=9										
36	1	29	222	645	910	645	222	29	1	2704
37	1	39	258	696	902	597	185	22		2700
38	2	50	302	738	894	543	156	15		2700
39	3	66	345	783	873	495	127	11		2703
40	5	82	394	816	851	441	104	7		2700
41	7	104	441	851	816	394	82	5		2700
42	11	127	495	873	783	345	66	3		2703
43	15	156	543	894	738	302	50	2		2700
44	22	185	597	902	696	258	39	1		2700

n=10											
45	1	41	392	1468	2724	2724	1468	392	41	1	9252
46	1	53	462	1603	2794	2630	1326	326	30		9225
47	2	71	544	1749	2860	2532	1197	273	22		9250
48	3	90	630	1887	2897	2414	1067	222	15		9225
49	5	116	731	2030	2929	2297	949	182	11		9250
50	7	145	833	2161	2934	2161	833	145	7		9226
51	11	182	949	2297	2929	2030	731	116	5		9250
52	15	222	1067	2414	2897	1887	630	90	3		9225
53	22	273	1197	2532	2860	1749	544	71	2		9250
54	30	326	1326	2630	2794	1603	462	53	1		9225

n=11												
55	1	55	669	3113	7352	9686	7352	3113	669	55	1	32066
56	1	74	792	3460	7708	9673	6964	2795	556	42		32065
57	2	95	939	3812	8060	9598	6580	2486	463	30		32065
58	3	124	1097	4191	8371	9499	6172	2208	378	22		32065
59	5	157	1281	4569	8672	9340	5776	1940	310	15		32065
60	7	200	1478	4970	8924	9160	5364	1703	248	11		32065
61	11	248	1703	5364	9160	8924	4970	1478	200	7		32065
62	15	310	1940	5776	9340	8672	4569	1281	157	5		32065
63	22	378	2208	6172	9499	8371	4191	1097	124	3		32065
64	30	463	2486	6580	9598	8060	3812	939	95	2		32065
65	42	556	2795	6964	9673	7708	3460	792	74	1		32065

n=12													
66	1	76	1109	6300	18320	30554	30554	18320	6300	109	76	1	112720
67	1	98	1317	7040	19496	31132	29849	17125	5597	921	56		112632
68	2	129	1564	7850	20696	31641	29087	15968	4962	766	42		112707
69	3	165	1838	8698	21863	32017	28218	14812	4368	628	30		112640
70	5	212	2156	9613	23034	32312	27302	13703	3836	515	22		112710
71	7	266	2505	10560	24152	32467	26295	12608	3342	415	15		112632
72	11	336	2907	11573	25261	32540	25261	11573	2907	336	11		112716
73	15	415	3342	12608	26295	32467	24152	10560	2505	266	7		112632
74	22	515	3836	13703	27302	32312	23034	9613	2156	212	5		112710
75	30	628	4368	14812	28218	32017	21863	8698	1838	165	3		112640
76	42	766	4962	15968	29087	31641	20696	7850	1564	129	2		112707
77	56	921	5597	17125	29849	31132	19496	7040	1317	98	1		112632

n=13														
78	1	100	1792	12197	42841	87691	110780	87691	42841	12197	1792	100	1	400024
79	1	132	2133	13734	46078	90814	110599	84484	39674	10806	1491	77		400023
80	2	170	2535	15383	49446	93728	110171	81118	36652	9521	1241	56		400023
81	3	220	2986	17193	52853	96517	109392	77703	33723	8372	1019	42		400023
82	5	278	3514	19117	56365	99061	108368	74176	30957	7314	838	30		400023
83	7	354	4100	21209	59888	101436	107009	70648	28292	6378	680	22		400023
84	11	441	4777	23420	63479	103530	105430	67044	25798	5526	552	15		400023
85	15	552	5526	25798	67044	105430	103530	63479	23420	4777	441	11		400023
86	22	680	6378	28292	70648	107009	101436	59888	21209	4100	354	7		400023
87	30	838	7314	30957	74176	108368	99061	56365	19117	3514	278	5		400023
88	42	1019	8372	33723	77703	109392	96517	52853	17193	2986	220	3		400023
89	56	1241	9521	36652	81118	110171	93728	49446	15383	2535	170	2		400023
90	77	1491	10806	39674	84484	110599	90814	46078	13734	2133	132	1		400023

n=14															
91	1	134	2838	22845	95121	233814	361677	361677	233814	95121	22845	2838	134	1	1432860
92	1	173	3377	25785	103090	244790	366851	355701	222738	87491	20150	2365	101		1432613
93	2	225	4014	29040	111485	255762	371365	349134	211751	80299	17735	1969	77		1432858
94	3	286	4735	32571	120195	266528	375017	341843	200763	73456	15536	1624	56		1432613
95	5	366	5577	36449	129311	277185	377961	334040	189953	67050	13580	1339	42		1432858
96	7	459	6526	40629	138704	287528	380001	325604	179229	60994	11811	1091	30		1432613
97	11	578	7621	45190	148464	297658	381301	316750	168759	55363	10251	890	22		1432858
98	15	717	8847	50074	158453	307363	381676	307363	158453	50074	8847	717	15		1432614
99	22	890	10251	55363	168759	316750	381301	297658	148464	45190	7621	578	11		1432858
100	30	1091	11811	60994	179229	325604	380001	287528	138704	40629	6526	459	7		1432613
101	42	1339	13580	67050	189953	334040	377961	277185	129311	36449	5577	366	5		1432858
102	56	1624	15536	73456	200763	341843	375017	266528	120195	32571	4735	286	3		1432613
103	77	1969	17735	80294	211751	349134	371365	255762	111485	29040	4014	225	2		1432858
104	101	2365	20150	87491	222738	355701	366851	244790	103090	25785	3377	173	1		1432613

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